

## MIDTERM ASSIGNMENT — PROMPT ENGINEERING

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AI Tool Used: ChatGPT (GPTn5.1)

### PART 1: PROMPT COMPARISON ANALYSIS

A. Domain & Task Domain: Engineering — Architecture Task: Explain the concept of “passive cooling in buildings” to first-year architecture students.

B. Version A — Vague Prompt:

"Explain passive cooling in buildings." AI Response (ChatGPT – GPTn5.1):

Passive cooling refers to design strategies that help buildings stay cool without mechanical systems like air conditioners. Examples include shading, natural ventilation, and using reflective materials.

C. Version B — Improved Prompt Improved Prompt:

“You are an architecture studio instructor. Explain the concept of passive cooling to first year architecture students using clear and simple language. Include one practical example from building design, and end with a short, three sentence summary. Avoid advanced terminology unless necessary.” AI Response (ChatGPT – GPTn5.1):

Passive cooling is a design approach that reduces indoor heat by using natural elements instead of machines. It relies on airflow, shading, and materials that prevent heat gain. For example, placing windows opposite each other allows cross ventilation, which naturally cools a space. Summary:

Passive cooling reduces heat without mechanical systems. It relies on air movement and shading. It lowers energy use and increases comfort.

D. Comparison Analysis

The first prompt resulted in a short definition that lacked depth, context, and practical architectural relevance. Because it was vague, the explanation felt more like a dictionary entry than guidance for architecture students. There was no real example or structure to help the reader connect the concept to actual building design.

The improved prompt was more effective because it guided the AI through role assignment, audience targeting, and structural constraints. By setting the role as an architecture instructor, the tone became more educational. Defining the audience pushed the explanation toward clarity and beginner friendly language. Requiring an example anchored the concept in real practice, and a structured summary reinforced understanding. These techniques made the output more detailed, relatable, and useful for architectural learning.

### PART 2: PRACTICAL PROMPT DESIGN COSTAR

C – Context:

Architecture students struggle to understand environmental design concepts like passive

cooling.

O – Objective:

Provide a clear explanation with examples and a mini quiz.

S – Style:

Straightforward, friendly, and practical.

T – Tone:

Supportive and professor-like.

A – Audience:

First year architecture students.

R – Response:

Give an explanation, two examples, and a three question quiz with answers.

FINAL COSTAR PROMPT:

“You are an architecture instructor. Explain passive cooling in simple, beginner friendly terms.

Include exactly two architectural examples and then provide a short three question quiz with answers. Keep the tone warm and clear, suitable for first year architecture students.” AI

Response (ChatGPT – GPTn5.1):

Passive cooling uses building design to reduce heat without machines. It focuses on airflow, shading, and material choices to keep interiors comfortable. Example 1: Deep roof overhangs that block summer sun but allow winter sun. Example 2: Courtyards that create natural air

movement. Quiz: (1) What is passive cooling? Answer: Cooling through design instead of machines. (2) Give one example. Answer:

Overhangs or courtyards. (3) Why is it important? Answer: It saves energy and increases comfort.

Refinement Added:

“Keep the entire response under 180 words, use bullet points for examples, and avoid technical jargon.

### PART 3: CRITICAL REFLECTION

A. Skill Progression At the start of this course, I used prompts casually without realizing how much influence they had over the final output. Over time, I began to understand that prompting is more like giving architectural instructions than just asking questions—clarity, constraints, and intention all shape the result. One breakthrough moment for me was when I saw how assigning a role completely changed the tone and accuracy of responses. Challenges came when balancing detail with simplicity; too many instructions made the prompt stiff, while too few led to vague answers. With practice, I learned to guide the AI the same way an architect guides a design: with purpose, clarity, and structure.

B. Practical Application As an architecture student, I see real opportunities to use prompt engineering in areas like conceptual design support, sustainability analysis, and generating study notes. When trying to understand topics such as daylighting, thermal mass, or passive cooling, prompt engineering can help simplify complex ideas. Techniques like audience targeting, structured outputs, and examples would allow me to generate tailored learning resources quickly. I also expect to use prompts for creating early-stage design concepts or exploring variations. The main challenge will be making sure the AI’s suggestions are realistic

and align with architectural principles, which means verifying information and not relying blindly on the output.

#### C. Limitations & Ethics

AI tools sometimes oversimplify design concepts or provide inaccurate technical information. They also lack awareness of local building codes, climate conditions, and cultural context. From an ethical standpoint, architects must ensure AI generated explanations or ideas do not mislead clients or promote unsafe designs. To use AI responsibly, I would always verify technical details, cite when AI is used, and ensure safety and accuracy remain the priority.

D. Goals for the Second Half I want to develop stronger skills in multistep prompts, design exploration, and sustainability focused prompting. I also want to learn how to use prompt engineering for architectural workflows like diagrams, analysis steps, and research summaries